Pay Predictor: Predicting Median Salary from Skills listed on Linkedin Job Postings

Khang Pham and Himnish Chhabra

1. Group Members

Our group consists of two members: Khang Pham and Himnish Chhabra.

Khang will focus on exploratory data analysis, data cleaning, statistical testing, and visual interpretation of results. This includes examining salary distributions, analyzing differences across industries and company sizes, and conducting hypothesis testing such as ANOVA and ANCOVA. Khang will also be responsible for presenting findings through clear visualizations and evaluating overall model performance.

Himnish will be responsible for implementing the predictive modeling pipeline. This includes preparing structured and textual features, building baseline and advanced machine learning models, and tuning model parameters. He will experiment with models such as linear regression, Support Vector Machines (SVMs), and neural networks.

Both members will collaborate on refining features, interpreting results, and writing the final report to ensure the project is cohesive and well-structured.

2. Motivation

The goal of this project is to investigate how job descriptions and required skills relate to salary levels in LinkedIn job postings. Specifically, we want to determine whether the textual content of a job posting can be used to predict its listed compensation.

Many job postings either provide wide salary ranges or omit salary information entirely. As research on pay transparency suggests (**?**), salary information remains inconsistent in online job markets. At the same time, companies compete for talent in an increasingly skill-driven economy. This raises an important question:

Can we measure the market value of professional skills directly from job posting data?

By analyzing job descriptions, industries, and company characteristics, we aim to identify which factors are most strongly associated with higher salaries. Our objective is to better understand how job requirements translate into compensation in today's labor market.

3. Literature Review

We will examine prior research on wage determination and salary prediction using online job postings. Previous studies have shown that specific keywords, required skills, and industry trends influence salary outcomes.

In particular, Chen and Autor (**?**) demonstrate that machine learning models can predict salary levels using textual features from job postings. Their work highlights the importance of language in understanding labor market value.

We will build on this literature by combining statistical hypothesis testing with predictive modeling to evaluate both group-level salary differences and individual salary predictions.

4. Data

We will use the LinkedIn Job Postings dataset (2023–2024) available on Kaggle. The dataset contains over 120,000 job listings and includes both structured and unstructured data. Structured variables include job title, company size, industry, location, minimum salary, maximum salary, and pay period. The dataset also includes unstructured text such as job descriptions and listed skills.

Before analysis, we will clean the dataset by removing extreme or unrealistic salary outliers and handling missing salary values. Because the dataset is relatively large, we may apply sampling techniques during model training to improve computational efficiency while maintaining representative results.

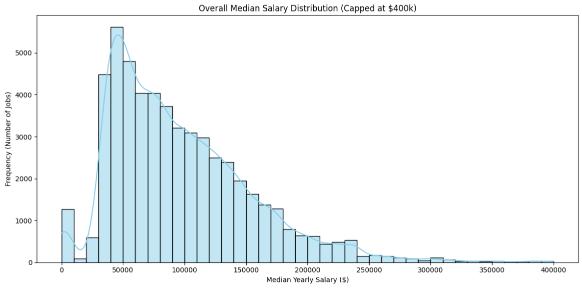

Figure 1: Median Salary Distribution Histogram

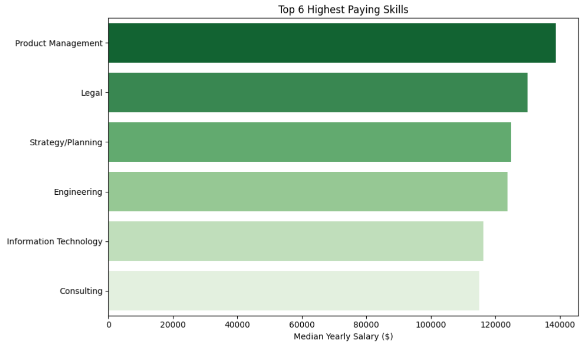

Figure 2: 6 Highest Paying Skills

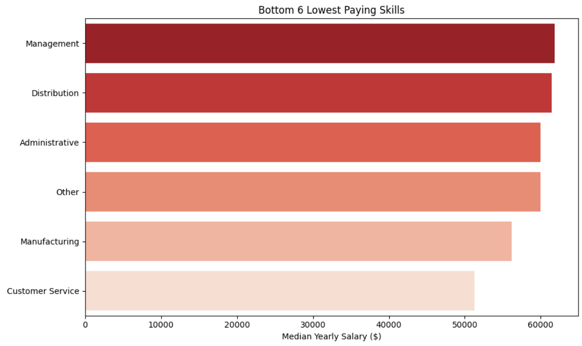

Figure 3: 6 Lowest Paying Skills

For exploratory data analysis, we will examine salary distributions overall and across categories such as industry, company size, and location. We will use visualizations including histograms and boxplots to identify patterns and variability in salary data.

To statistically test salary differences across groups, we will conduct hypothesis testing using:

- **ANOVA (Analysis of Variance)** – a statistical test used to determine whether the average salary differs significantly across multiple groups, such as industries.

- **ANCOVA (Analysis of Covariance)** – similar to ANOVA, but it controls for additional variables (such as company size or location) to determine whether salary differences remain significant after accounting for those factors.

These tests will help us determine whether observed salary differences are statistically meaningful rather than due to random variation.

5. Approach

Our approach combines statistical analysis with machine learning techniques.

First, we will preprocess the textual data by cleaning job descriptions and converting them into numerical features using methods such as TF-IDF or word embeddings. This allows the models to interpret text information mathematically.

Next, we will build baseline models using linear regression. Linear regression provides a simple and interpretable starting point for predicting salary based on structured variables (industry, company size, location) and text-based features.

After establishing a baseline, we will experiment with more advanced models such as Support Vector Machines (SVMs) and neural networks. These models can capture more complex and nonlinear relationships between job descriptions and compensation.

By comparing simple and advanced models, we aim to determine whether salary prediction improves significantly when using more flexible modeling techniques.

6. Evaluation Metrics

We will evaluate our results both qualitatively and quantitatively.

Qualitatively, we will produce visualizations such as salary distribution plots, industry comparisons, and the predicted-versus-actual salary scatterplot. These will help us interpret patterns and understand model behavior.

Quantitatively, because salary prediction is a regression problem, we will use:

- **Mean Squared Error (MSE)** – measures the average squared difference between predicted and actual salaries. Lower values indicate better performance.

- **Root Mean Squared Error (RMSE)** – similar to MSE but expressed in salary units (dollars), making it easier to interpret the prediction error.

- **R-squared** (R^2) – measures the proportion of salary variation explained by the model. Higher values indicate a stronger explanatory power.

We will also use cross-validation to ensure that our results generalize beyond the training data.

For ANOVA and ANCOVA, we will report p-values and F-statistics to determine whether salary differences between groups are statistically significant.

If advanced models substantially reduce prediction error compared to linear regression, this would suggest that salary depends on complex patterns within job descriptions.

7. Pre-Processing

First, we standardize the target variable, `med_salary`, by converting all hourly and monthly rates into a consistent yearly figure using conversion factors of $\times 2{,}080$ for hourly rates (40 h/week $\times$ 52 weeks) and $\times 12$ for monthly rates. Where the median salary is missing, we impute it from the mean of the available minimum and maximum bounds. Records with no salary signal whatsoever are dropped.

We then perform a high-integrity merge between `salaries.csv`, `postings.csv`, and `job_industries.csv` (joined with `industries.csv`) using `job_id` as the primary key. Extreme salary values below $15,000 or above $600,000 per year are removed as implausible outliers, leaving **35,379 job postings** for modeling. The final merged dataset covers all seven work types, multiple experience levels, and hundreds of industry categories.

For the unstructured text, we normalize job descriptions by stripping HTML tags with a regular expression and removing all non-alphabetic characters. The cleaned text is then lower-cased and reduced to single-space separation, producing a corpus ready for vectorization. Categorical variables (`work_type`, `formatted_experience_level`, `industry_name`) are label-encoded via one-hot encoding.

8. Exploratory Data Analysis

Our exploratory analysis focuses on identifying the primary structural and skill-based drivers of compensation. This phase involved analyzing salary distributions across work types, skill volumes, and specific technical domains to uncover the underlying patterns in the LinkedIn dataset.

Overall Salary Distribution

As shown in Figure **??**, the distribution of annual median salaries is strongly right-skewed with a mode near $50,000–$60,000, consistent with the large proportion of full-time, entry-to-mid-level U.S. positions. A long tail extends toward $400,000, corresponding to executive and specialized technical roles. The presence of this skew motivated our use of a salary cap at $400,000 in exploratory plots and a broader $600,000 ceiling for modeling (to include senior executive postings while discarding clearly erroneous data entry). The salary distribution also highlights that approximately 8% of postings fall below $20,000, a cluster that largely represents part-time and internship roles already distinguished by `work_type`.

a. Work Type Spread

Shown in Figure **??**, there is significant variance in salary across different employment categories. Full-time and Contract roles exhibit the highest median salaries and the widest ranges, with several postings reaching the $400,000 cap. In contrast, Internship, Part-time, and Volunteer roles cluster at the lower end of the spectrum, usually below $75,000, suggesting that `work_type` is a critical categorical feature for our regression models.

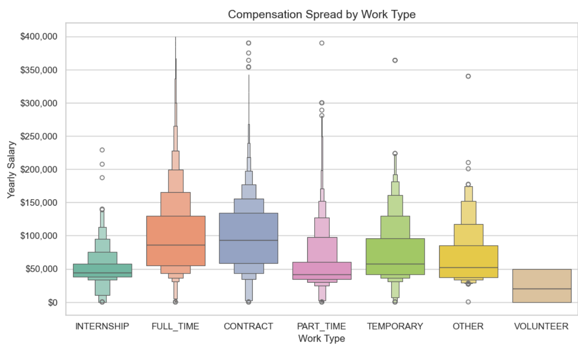

Figure 4: Compensation Spread by Work Type

b. Skill Requirement Volume

Since our dataset contains jobs that listed multiple skills, we plotted the median salary for jobs listing multiple unique skills. Interestingly, we see that requiring 2 unique skills averages a lower median salary than having 1 and 3. This "threshold effect" suggests that jobs requiring a specific trifecta of skills are valued significantly higher by employers, justifying our plan to use non-linear models like Neural Networks to capture these complex interactions.

Figure 5: Impact of Skill Requirement Volume on Salary

c. Salary Distribution for Demanded Skills

We gathered the 5 most frequent industry categories from the dataset and graphed a box plot. We see that Engineering and Information Technology contains the highest median salary listing at above \$100,000. In contrast, Management and Sales roles show lower median pay and narrower interquartile ranges. Manufacturing roles exhibit the lowest median pay among the top five.

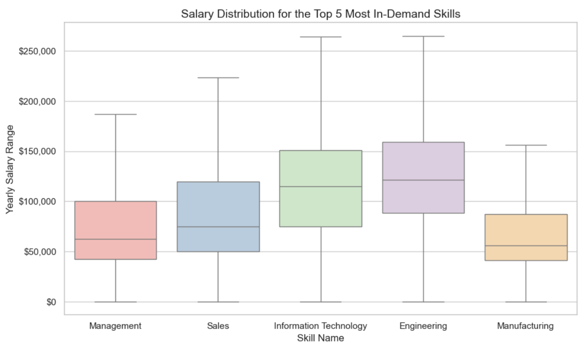

Figure 6: Salary Distribution for Most In-Demand Skills

d. Key Observations from EDA

Several structural patterns emerged that directly informed our modeling decisions. First, `work_type` is the single strongest categorical predictor: full-time and contract roles command median salaries roughly 2× higher than internships and part-time roles. Second, the non-monotonic relationship between skill count and salary (Figure **??**) suggests interaction effects that linear models may struggle to capture, motivating the use of non-linear models in future work. Third, the high-paying skills bar chart (output2) shows that niche technical qualifications such as quantum computing and chip architecture carry strong premiums, while low-paying postings (output3) tend to emphasize generic soft skills. This asymmetry validates the use of TF-IDF text features: salary signal is concentrated in specific vocabulary rather than general language patterns.

9. Statistical Analysis

To validate our hypotheses, we conducted two primary statistical tests:

a. One-Way ANOVA

We tested if heart rate changes significantly based on required skills. We performed a One-Way ANOVA test on skills for job listings.

Results: The test resulted in an F-statistic of 4.3275 and a p-value of 0.001685. The extremely small p-value indicates that the differences in salary across these skill categories are statistically significant, justifying their use as key features in our predictive models.

b. Two-Sample T-Test

We conducted a t-test to compare the average salaries of roles requiring specific high-value skills (e.g., Information Technology) against the general population.

Results: The test yielded a p-value of 0.0144. Since the p-value is below the 0.05 threshold, we reject the null hypothesis and conclude that the targeted skill significantly influences salary levels.

10. Model Training and Evaluation

a. Feature Representation

We built two versions of the model to see whether job description text improves salary prediction:

- **Model A (structured only):** Uses categorical job information – work type, experience level, and industry – converted into numerical format.

- **Model B (structured + text):** Everything in Model A, plus text features from job descriptions. We used TF-IDF, which turns words into numbers based on how often they appear and how unique they are. We kept the top 5,000 words/phrases to capture the most useful vocabulary.

b. Model

Both models use Ridge Regression, a form of linear regression that adds a penalty to prevent the model from over-relying on any single word or feature – especially important when dealing with thousands of text features. We split the data 80/20: 28,303 jobs for training and 7,076 for testing.

c. Results

Table 1 shows how well each model predicted salary on the test set.

Model	RMSE ($)	R^2
Model A: Structured only	47,754	0.320
Model B: Structured + TF-IDF	36,843	0.595
5-Fold CV (Model B): $R^2 = 0.594 \pm 0.015$, $RMSE = \$36{,}998 \pm \$1{,}432$		

Table 1: Test-set performance of Ridge Regression models.

Adding text features cut prediction error by **$10,911** (23%) and raised R^2 from 0.320 to 0.595, meaning the model now explains roughly 60% of salary variation. Cross-validation scores closely matched these results, showing the model generalizes well to unseen data.

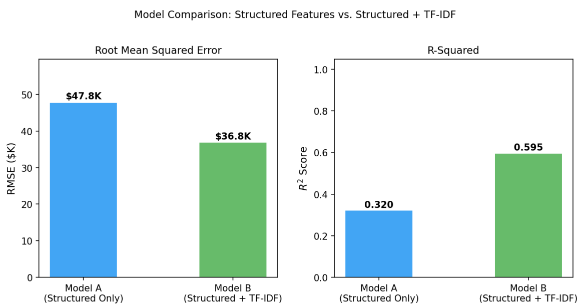

Figure 7: Model comparison: RMSE and R^2 for Model A vs. Model B.

d. What the Words Tell Us

Figure **??** shows which words in job descriptions are most associated with higher or lower salaries. Words linked to high pay tend to be technical terms from fields like machine learning, finance, and cloud computing. Words linked to lower pay tend to come from service-sector jobs (e.g., shift scheduling, customer service). This aligns with our EDA findings and confirms that salary information is encoded in specific vocabulary, not just general language.

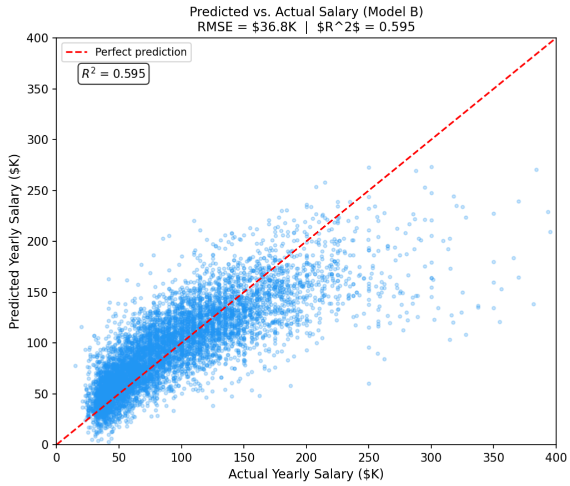

Figure 8: Predicted vs. actual salary on the test set (Model B). The dashed red line represents perfect prediction.

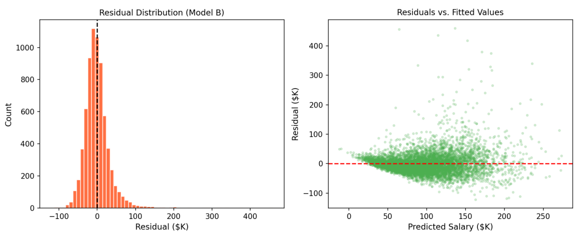

Figure 9: Residual distribution and residuals vs. fitted values for Model B.

e. Residual Analysis

The residual plot (Figure **??**) shows how far off our predictions were. Errors are roughly centered at zero, but the model tends to under-predict the highest salaries – a common issue with linear models when salaries are heavily skewed. Prediction error also grows at higher salary levels, suggesting that future work with non-linear models or a log-transformed salary target could improve accuracy for senior roles.

f. Challenges

- **Limited salary data.** Only 35,379 of 120,000+ postings had usable salary info after cleaning – a 71% reduction – limiting how broadly our model applies.
- **Many features, limited data.** With 5,000 text features and ~28k samples, regularization was essential to prevent overfitting.
- **Growing prediction error.** The model struggles more with high-salary roles; future work

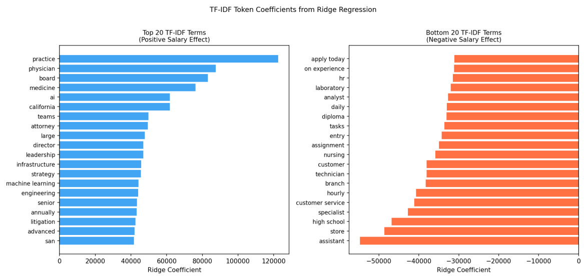

Figure 10: Top 20 TF-IDF tokens with the largest positive (left) and negative (right) Ridge coefficients.

will explore log-transformation or non-linear models.

- **Missing experience levels.** About 30% of postings had no experience level listed and were labeled "Unknown," which may weaken that feature's usefulness.

- **Too many industry categories.** Hundreds of fine-grained industries means rare ones have very few examples. Grouping small industries into an "Other" category may help in future iterations.

References

Shanshan Chen and David Autor. *Words Matter: AI Can Predict Salaries Based on the Text of Online Job Postings*. Stanford Digital Economy Lab Research Brief, 2024.

David Arnold, Simon Quach, and Bledi Taska. *The Impact of Pay Transparency in Job Postings on the Labor Market*. NBER Working Paper Series, w34480:1–48, 2025.